

Barriers and drivers for protein transition among older Australians: A cross-sectional study

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INTRODUCTION

A partial shift from animal derived protein foods to plant-based protein foods (protein transition) may support healthy ageing and environmental sustainability. The factors influencing protein food choices among older adults are not well understood. This study investigates the barriers and drivers for protein transition among older Australians, using the COM-B (Capability, Opportunity, Motivation - Behaviour) model.

METHODS

A cross-sectional online survey was completed by 501 independently living Australians aged 60–75 years. Multiple linear regressions examined demographic, psychosocial, and practical factors associated with plant-protein, animal-protein and vegetable intake, and willingness to reduce animal-protein intake. Predictors were grouped across COM-B. Examples of predictors include cooking confidence (capability) food affordability (opportunity) and dietary habits (motivation). COM-B domains were then entered sequentially into hierarchical regression models.

RESULTS AND DISCUSSION

Key associations were as follows:

Plant-protein intake: Greater intake was associated with stronger plant-protein dietary habits ($\beta = 0.18, p = 0.009$), greater cooking confidence ($\beta = 0.18, p = 0.010$) and a greater sense of responsibility for personal health ($\beta = 0.11, p = 0.024$). The model explained 27.0% of the variance in plant-protein intake.

Animal-protein intake: Greater intake was associated with being a man ($\beta = -0.12, p = 0.012$), having a household income of \$80,000–\$119,999 ($\beta = 0.12, p = 0.030$ relative to $< \$40,000$), weaker beliefs in the health benefits of reducing animal-protein foods ($\beta = -0.16, p = 0.002$), and stronger beliefs about the health risks of reducing them ($\beta = 0.17, p = 0.001$). The model explained 17.5% of the variance in animal-protein intake.

Vegetable intake: Greater intake was associated with being a woman ($\beta = 0.15, p = 0.001$), university education ($\beta = 0.12, p = 0.015$), and a greater sense of responsibility for health ($\beta = 0.17, p = 0.001$). The model explained 8.2% of the variance.

Willingness to reduce animal-protein intake: Greater willingness was associated with lower meat attachment ($\beta = -0.36, p < 0.001$), stronger beliefs in the health benefits of reducing animal-protein foods ($\beta = 0.33, p < 0.001$), and unexpectedly, less physical

difficulty eating animal-protein foods ($\beta = -0.11, p = 0.007$). The model explained 46.4% of the variance.

When predictors were grouped by COM-B component, capability and motivation contributed most consistently, while opportunity-related factors may have a more limited independent role (Figure 1).

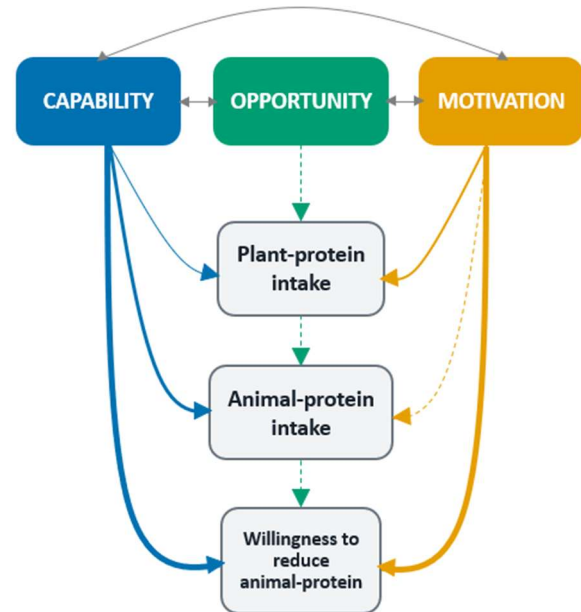


Figure 1 COM-B contributions to protein related outcomes. Solid and dashed arrows indicate significant and non-significant independent contributions respectively. Grey arrows indicate correlations observed between domains.

CONCLUSIONS

These findings identify potentially modifiable factors relevant to protein transition among older Australians. Strategies could target capability by building cooking confidence of plant-protein foods and addressing perceptions of the health benefits and risks of reducing animal-derived foods; and motivation by supporting habitual plant-protein consumption, addressing meat attachment, and leveraging older adults' sense of personal responsibility for their health.

ACKNOWLEDGEMENTS

This project has been funded by Hort Innovation, using the vegetable research and development levy and contributions from the Australian Government. Hort Innovation is the grower-owned, not-for-profit research and development corporation for Australian horticulture. Project Number: VG24004